

ENM 5310 — Quiz 1 · VERSION A

10 minutes. Closed book, closed notes, no devices. One best answer per question. On your bubble sheet: write and bubble your **name** and **Penn ID**, and bubble **Version A**. Pencil only. Each question has five bubbles (A–E) — **ignore E**. No penalty for wrong answers, so answer everything.

1. You estimate $\mathbb{E}[f(X)]$ by a sample average over N i.i.d. draws. You currently have $\text{RMSE} = 10^{-2}$ and want 10^{-3} . Assuming $\mathbb{V}[f(X)] < \infty$, roughly how many samples do you need, and how does the answer change if X lives in $\mathbb{R}^{100}$ rather than $\mathbb{R}$?
- (A) $10\times$ more samples; the requirement grows exponentially with dimension.
 - (B) $1000\times$ more samples; the rate degrades as $N^{-1/3}$ in high dimensions.
 - (C) $100\times$ more samples; the requirement is unchanged by dimension.
 - (D) $100\times$ more samples; in $\mathbb{R}^{100}$ the required N grows by a further factor exponential in the dimension.
2. Data are generated as $y = f(x) + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, 0.25)$ independent of x . Minimizing over **all** measurable functions g , the smallest achievable value of $\mathbb{E}[(y - g(x))^2]$ on new data is:
- (A) 0, achieved by $g = f$ given enough data and enough capacity.
 - (B) 0.25, achieved by $g = f$; no model can do better.
 - (C) 0.5, the standard deviation of the noise.
 - (D) It cannot be determined without specifying the model class for g .
3. Let X have a distribution symmetric about 0 with finite variance, and let $Y = X^2$. Which statement is correct?
- (A) $\rho_{XY} = 0$, and therefore X and Y are independent.
 - (B) $\rho_{XY} \neq 0$, because Y is a deterministic function of X and correlation detects any dependence.
 - (C) $\rho_{XY} = 0$, but X and Y are dependent; zero correlation would imply independence only if (X, Y) were jointly Gaussian.
 - (D) ρ_{XY} cannot be determined without specifying the distribution of X .
4. A screening test has sensitivity $p(+ \mid \text{defective}) = 0.9$ and specificity $p(- \mid \text{sound}) = 0.9$. In the production lot, 1% of parts are defective. A randomly selected part tests positive. The probability that it is actually defective is closest to:
- (A) 1% (B) 8% (C) 50% (D) 90%
5. Let $X \sim \text{Unif}(0, 1)$ and $Y = e^X$. The density of Y is:
- (A) $p_Y(y) = 1/y$ for $y \in [1, e]$, zero otherwise.
 - (B) $p_Y(y) = 1$ for $y \in [1, e]$, zero otherwise.
 - (C) $p_Y(y) = y$ for $y \in [1, e]$, zero otherwise.
 - (D) $p_Y(y) = e^{-y}$ for $y \geq 0$.
6. Let q be the data distribution and p a model. Which statement about $\text{KL}(q \parallel p) = \mathbb{E}_q[\log(q(X)/p(X))]$ is correct?
- (A) It is symmetric in its arguments and satisfies the triangle inequality, so it is a distance between distributions.
 - (B) It can be negative when p has heavier tails than q , since the log-ratio is then negative over much of the support.
 - (C) It equals the entropy $\mathbb{H}[q]$ whenever p is uniform on the support of q .
 - (D) It is non-negative, zero only when $p = q$, and measures the excess average surprise incurred by modeling data from q using p .
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END — confirm you bubbled **Version A**.

ENM 5310 — Quiz 1 · VERSION B

10 minutes. Closed book, closed notes, no devices. One best answer per question. On your bubble sheet: write and bubble your **name** and **Penn ID**, and bubble **Version B**. Pencil only. Each question has five bubbles (A–E) — **ignore E**. No penalty for wrong answers, so answer everything.

1. You estimate $\mathbb{E}[f(X)]$ by a sample average over N i.i.d. draws. You currently have $\text{RMSE} = 10^{-2}$ and want 10^{-3} . Assuming $\mathbb{V}[f(X)] < \infty$, roughly how many samples do you need, and how does the answer change if X lives in $\mathbb{R}^{100}$ rather than $\mathbb{R}$?
- (A) $1000\times$ more samples; the rate degrades as $N^{-1/3}$ in high dimensions.
 - (B) $100\times$ more samples; in $\mathbb{R}^{100}$ the required N grows by a further factor exponential in the dimension.
 - (C) $10\times$ more samples; the requirement grows exponentially with dimension.
 - (D) $100\times$ more samples; the requirement is unchanged by dimension.
2. Data are generated as $y = f(x) + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, 0.25)$ independent of x . Minimizing over **all** measurable functions g , the smallest achievable value of $\mathbb{E}[(y - g(x))^2]$ on new data is:
- (A) 0.5, the standard deviation of the noise.
 - (B) It cannot be determined without specifying the model class for g .
 - (C) 0.25, achieved by $g = f$; no model can do better.
 - (D) 0, achieved by $g = f$ given enough data and enough capacity.
3. Let X have a distribution symmetric about 0 with finite variance, and let $Y = X^2$. Which statement is correct?
- (A) $\rho_{XY} \neq 0$, because Y is a deterministic function of X and correlation detects any dependence.
 - (B) $\rho_{XY} = 0$, but X and Y are dependent; zero correlation would imply independence only if (X, Y) were jointly Gaussian.
 - (C) ρ_{XY} cannot be determined without specifying the distribution of X .
 - (D) $\rho_{XY} = 0$, and therefore X and Y are independent.
4. A screening test has sensitivity $p(+ \mid \text{defective}) = 0.9$ and specificity $p(- \mid \text{sound}) = 0.9$. In the production lot, 1% of parts are defective. A randomly selected part tests positive. The probability that it is actually defective is closest to:
- (A) 90% (B) 50% (C) 8% (D) 1%
5. Let $X \sim \text{Unif}(0, 1)$ and $Y = e^X$. The density of Y is:
- (A) $p_Y(y) = e^{-y}$ for $y \geq 0$.
 - (B) $p_Y(y) = 1$ for $y \in [1, e]$, zero otherwise.
 - (C) $p_Y(y) = y$ for $y \in [1, e]$, zero otherwise.
 - (D) $p_Y(y) = 1/y$ for $y \in [1, e]$, zero otherwise.
6. Let q be the data distribution and p a model. Which statement about $\text{KL}(q \parallel p) = \mathbb{E}_q[\log(q(X)/p(X))]$ is correct?
- (A) It is non-negative, zero only when $p = q$, and measures the excess average surprise incurred by modeling data from q using p .
 - (B) It equals the entropy $\mathbb{H}[q]$ whenever p is uniform on the support of q .
 - (C) It is symmetric in its arguments and satisfies the triangle inequality, so it is a distance between distributions.
 - (D) It can be negative when p has heavier tails than q , since the log-ratio is then negative over much of the support.
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END — confirm you bubbled **Version B**.